CH8

(a)

```
vacation <- read_csv("vacation.csv")
```

```
## Rows: 200 Columns: 4
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## dbl (4): miles, income, age, kids
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
model <- lm(miles ~ income + age + kids, data = vacation)

summary(model)
```

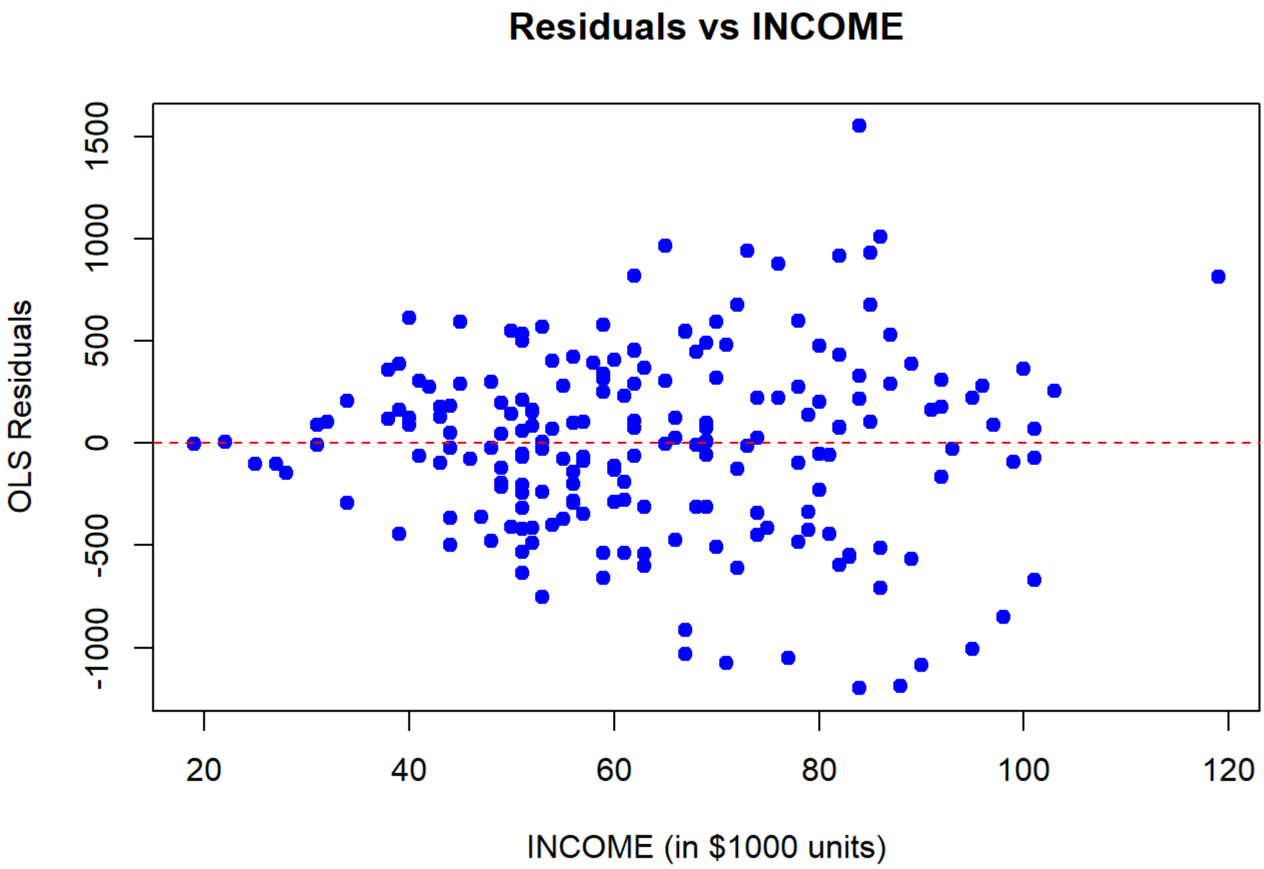
```
##
## Call:
## lm(formula = miles ~ income + age + kids, data = vacation)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1198.14  -295.31    17.98   287.54  1549.41
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -391.548    169.775  -2.306   0.0221 *
## income        14.201      1.800   7.889 2.10e-13 ***
## age           15.741      3.757   4.189 4.23e-05 ***
## kids         -81.826     27.130  -3.016   0.0029 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 452.3 on 196 degrees of freedom
## Multiple R-squared:  0.3406, Adjusted R-squared:  0.3305
## F-statistic: 33.75 on 3 and 196 DF,  p-value: < 2.2e-16
```

```
ci_kids <- confint(model, parm = "kids", level = 0.95)
ci_kids
```

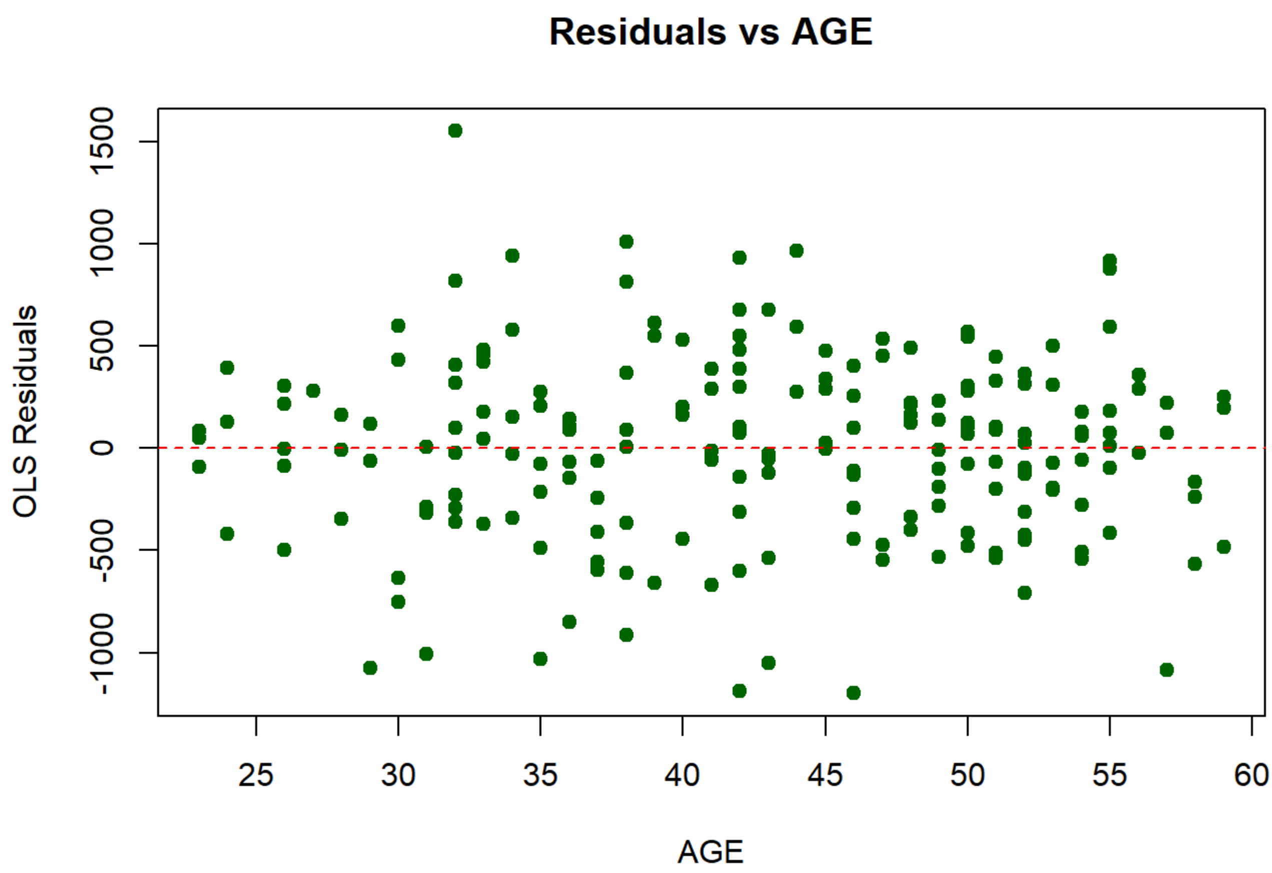
```
##          2.5 %    97.5 %
## kids -135.3298 -28.32302
```

(b)

```r
residuals_ols <- resid(model)

plot(vacation$income, residuals_ols,
     xlab = "INCOME (in $1000 units)",
     ylab = "OLS Residuals",
     main = "Residuals vs INCOME",
     pch = 19, col = "blue")

abline(h = 0, col = "red", lty = 2)
```




```r
plot(vacation$age, residuals_ols,
     xlab = "AGE",
     ylab = "OLS Residuals",
     main = "Residuals vs AGE",
     pch = 19, col = "darkgreen")

abline(h = 0, col = "red", lty = 2)
```



收入增加殘差

變異遞增，年齡則無明顯趨勢

(c)

```
vacation_sorted <- vacation[order(vacation$income), ]

n <- nrow(vacation_sorted)

data1 <- vacation_sorted[1:90, ]
data2 <- vacation_sorted[(n - 90 + 1):n, ]

model1 <- lm(miles ~ income + age + kids, data = data1)
model2 <- lm(miles ~ income + age + kids, data = data2)

rss1 <- sum(residuals(model1)^2)
rss2 <- sum(residuals(model2)^2)

k <- length(coef(model1))

df1 <- nrow(data1) - k
df2 <- nrow(data2) - k

F_stat <- (rss2 / df2) / (rss1 / df1)

p_value <- pf(F_stat, df1 = df2, df2 = df1, lower.tail = FALSE)

cat("--- Goldfeld-Quandt Test Results ---\n")
```

```
## --- Goldfeld-Quandt Test Results ---
```

```
cat("RSS1, lower income group:", rss1, "\n")
```

```
## RSS1, lower income group: 8750040
```

```
cat("RSS2, higher income group:", rss2, "\n")
```

```
## RSS2, higher income group: 27160654
```

```
cat("F-statistic:", F_stat, "\n")
```

```
## F-statistic: 3.104061
```

```
cat("Degrees of Freedom:", df2, "and", df1, "\n")
```

```
## Degrees of Freedom: 86 and 86
```

```
cat("p-value:", p_value, "\n")
```

```
## p-value: 1.64001e-07
```

由於

$p=1.64\times10^{-7}<0.05$

因此在 5% 顯著水準下，拒絕虛無假設 H0

表示有充分證據顯示高收入家庭與低收入家庭的誤差變異數並不相同，且高收入家庭的誤差變異數顯著較大，因此模型存在異質變異數（heteroskedasticity）。

(d)

```
cov_matrix <- vcovHC(model, type = "HC1")

robust_results <- coeftest(model, vcov = cov_matrix)
robust_results
```

```
##
## t test of coefficients:
##
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept) -391.5480   142.6548 -2.7447 0.0066190 **
## income        14.2013     1.9389  7.3246 6.083e-12 ***
## age           15.7409     3.9657  3.9692 0.0001011 ***
## kids         -81.8264    29.1544 -2.8067 0.0055112 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
coef_kids <- coef(model)["kids"]
se_kids_robust <- sqrt(diag(cov_matrix))["kids"]

n <- nrow(model$model)
k <- length(coef(model))

t_critical <- qt(0.975, df = n - k)

lower_bound <- coef_kids - t_critical * se_kids_robust
upper_bound <- coef_kids + t_critical * se_kids_robust

cat("--- 95% Robust Interval Estimate for KIDS ---\n")
```

```
## --- 95% Robust Interval Estimate for KIDS ---
```

```r
cat(sprintf("Robust CI for kids: [%.4f, %.4f]\n", lower_bound, upper_bound))
```

```
## Robust CI for kids: [-139.3230, -24.3299]
```

(e)

```r
gls_model <- lm(miles ~ income + age + kids,
                weights = 1 / (income^2),
                data = vacation)

summary(gls_model)
```

```
##
## Call:
## lm(formula = miles ~ income + age + kids, data = vacation, weights = 1/(income^2))
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1907  -4.9555   0.2488   4.3832  18.5462
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -424.996    121.444  -3.500 0.000577 ***
## income        13.947      1.481   9.420  < 2e-16 ***
## age           16.717      3.025   5.527 1.03e-07 ***
## kids         -76.806     21.848  -3.515 0.000545 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.765 on 196 degrees of freedom
## Multiple R-squared:  0.4573, Adjusted R-squared:  0.449
## F-statistic: 55.06 on 3 and 196 DF,  p-value: < 2.2e-16
```

```r
gls_ci <- confint(gls_model, parm = "kids", level = 0.95)

cat("--- Conventional GLS 95% Confidence Interval for KIDS ---\n")
```

```
## --- Conventional GLS 95% Confidence Interval for KIDS ---
```

```
gls_ci
```

```
##          2.5 %    97.5 %
## kids -119.8945 -33.71808
```

```
cov_matrix_gls <- vcovHC(gls_model, type = "HC1")

robust_gls_results <- coeftest(gls_model, vcov = cov_matrix_gls)
robust_gls_results
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value  Pr(>|t|)
## (Intercept) -424.9962    95.8035 -4.4361 1.526e-05 ***
## income        13.9473     1.3470 10.3545 < 2.2e-16 ***
## age           16.7175     2.7974  5.9761 1.061e-08 ***
## kids         -76.8063    22.6186 -3.3957 0.0008286 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coef_kids_gls <- coef(gls_model)["kids"]
se_kids_gls_robust <- sqrt(diag(cov_matrix_gls))["kids"]

n <- nrow(gls_model$model)
k <- length(coef(gls_model))

t_critical_gls <- qt(0.975, df = n - k)

lower_gls_robust <- coef_kids_gls - t_critical_gls * se_kids_gls_robust
upper_gls_robust <- coef_kids_gls + t_critical_gls * se_kids_gls_robust

cat("--- Robust GLS 95% Confidence Interval for KIDS ---\n")
```

```
## --- Robust GLS 95% Confidence Interval for KIDS ---
```

```
cat(sprintf("Robust GLS CI for kids: [%.4f, %.4f]\n",
            lower_gls_robust, upper_gls_robust))
```

```
## Robust GLS CI for kids: [-121.4134, -32.1992]
```

代表使用 White heteroskedasticity-robust standard errors 後，KIDS 的標準誤略為增加，因此信賴區間稍微擴大。